

Humana JourneyAI

Feature and UI Build Guide for the hackathon demo

Goal: reduce preventable member and provider contacts by turning claims, ROI status, and care gaps into clear explanations, next steps, and representative-ready summaries.

Quick Architecture

Area	What it does	Primary screens
Agentic workflow	Runs before a call. Claim, ROI, and Care Gap agents read structured data, then the Intervention Agent generates explanations, recommendations, and provider summaries.	Member Dashboard, Provider Action Center, Agent Activity Panel
Claim Journey Dashboard	Shows the current issue, owner, next step, estimated resolution, ROI status, and care-gap prompts so members do not need to call for basic visibility.	Member Dashboard
Call assist workflow	Runs when a member or provider still calls. The Escalation Agent loads the AI-generated summary so the representative can start resolving the issue instead of investigating from scratch.	Representative View, Escalation Screen

Feature Inventory by Step

Step / Agent	Features to build	Description / behavior	Potential UI screens
Claim Agent	- Claim status lookup - Denial reason lookup - Coverage rule lookup - Prior authorization check - Owner identification - Estimated resolution - Plain-English explanation	Reads claims and coverage_rules. Converts confusing status or denial information into a clear issue, owner, and next step. Example: Missing Prior Authorization, Owner: Provider, Next Step: Submit authorization.	Member Dashboard Claim Timeline AI Explanation Panel Provider Action Center
ROI Agent	- ROI authorization validation - Family-member authorization check - Missing ROI detection - ROI guidance generation - Authorization status display	Reads roi_authorizations. Prevents dead-end calls by checking whether sensitive claim information can be discussed. If missing, it explains that an ROI form is required.	ROI Status Widget ROI Screen Representative View
Care Gap Agent	- Care gap detection - Overdue screening detection - Wellness visit detection - Appointment lookup - Preventive care recommendation - Optional Stars impact indicator	Reads care_gaps and appointment_slots. Identifies preventive care opportunities and suggests a next step, such as an available appointment slot.	Care Gap Widget Care Gap Screen Member Dashboard Sidebar
Intervention Agent	- Decide next action - Generate member explanation - Generate provider summary - Generate ROI instructions - Generate care gap recommendation - Update dashboard - Prioritize unresolved issues	Turns agent findings into a usable action. This is the bridge between backend intelligence and the UI. Outputs explanations, summaries, and recommendations.	AI Explanation Panel Provider Summary Agent Activity Feed Dashboard Updates
Claim Journey Dashboard	- Claim timeline - Current issue card - Owner card - Next steps card - Estimated resolution card - ROI status widget - Care gap widget - Contact risk widget	The main member-facing screen. It provides Amazon-style visibility: what happened, why it happened, who owns it, and what happens next.	Member Dashboard

Step / Agent	Features to build	Description / behavior	Potential UI screens
Provider Summary Generator	- Provider action report - Claim issue summary - Missing authorization summary - Required action - Timeline summary - View full summary button	Creates a provider-facing action summary when the provider is the blocker. It does not claim to send real messages; it generates the report that could appear in a provider portal or rep view.	Provider Action Center Provider Summary Modal Representative View
Escalation Agent	- Confidence checking - Appeal detection - Compliance flag awareness - Human escalation decision - AI case summary - Next-step recommendation	Used when the issue should not be auto-handled. It attaches root cause, status, actions already taken, and suggested resolution to the case record.	Escalation Screen Representative View
Representative View	- Last name / member ID lookup - AI case summary - Root cause summary - Actions completed - Suggested resolution - Claim timeline	Used during the call. The rep sees the issue instantly and can begin solving it without re-investigating the claim from scratch.	Call Center / Representative Screen

Potential Screens to Mock Up

Screen	Purpose	Core UI components	Demo moment
Member Dashboard	The hero screen. Shows the member what is happening before they call.	Claim timeline, current issue card, owner, next step, estimated resolution, AI explanation, contact risk widget, ROI status, care-gap widget.	Show MRI claim stuck at Miss PA, explain why, and show no action needed from member.
AI Explanation Panel	Turns backend data into plain language.	Question title, explanation text, next steps, supportive note, confidence badge.	Click Explain This and show the simple explanation of the claim delay.
Agent Activity Feed	Makes the demo feel truly agentic.	Claim Agent found Missing PA, ROI Agent verified authorization, Care Gap Agent found overdue screening, Intervention Agent generated summary.	Show judges that multiple agents contributed to the final dashboard state.
Provider Action Center	Shows the provider what action is required.	Action Required card, claim, issue, priority, required action, claim impact, member impact, view full summary button.	Move from member view to provider view and show Submit PA as the required action.
Care Gap Screen	Shows proactive health opportunities beyond claims.	Care gap list, overdue status, appointment recommendation, health impact explanation, schedule CTA mockup.	Show Diabetic Eye Exam overdue with a Tuesday 2 PM appointment opportunity.
ROI Screen	Prevents authorization-related dead-end calls.	Authorization status, required action, ROI explanation, submit ROI form CTA mockup.	Show ROI missing and explain why claim information cannot be discussed yet.
Escalation Screen	Shows when AI hands off safely.	Escalation reason, confidence indicator, AI summary, root cause, actions taken, recommended next step.	Show confidence is too low or appeal/compliance risk exists, then route to human.
Representative View	Improves unavoidable calls.	Member lookup, AI case summary, claim timeline, actions completed, suggested resolution panel.	Caller says last name/member ID; summary loads immediately for the rep.

Mockup Wireframe Ideas

Member Dashboard layout	Top: My Health Journey header. Left: Current Issue card. Right: AI Explanation panel. Middle: Claim Journey Timeline. Bottom: ROI Status, Contact Risk Score, and Care Gap Recommendations.
Provider Action Center layout	Top: Provider Action Center header. Main card: Action Required. Below: Required Action, Claim Impact, Member Impact. CTA: View Full Provider Summary.
Representative View layout	Top: caller lookup by last name/member ID. Main panel: AI Case Summary. Side panel: Claim Timeline and Suggested Resolution. Bottom: Actions Completed checklist.
Agent Activity Feed layout	Narrow right-side panel listing each agent and its latest output: Claim Agent, ROI Agent, Care Gap Agent, Intervention Agent, Escalation Agent. Include confidence badge.

Must-Have Demo Features

Priority	Feature	Why it matters	Screen
Must	Claim Timeline	This is the Amazon-tracking analogy and the easiest feature for judges to remember.	Member Dashboard
Must	AI Explanation Panel	Shows the value of AI translating complex claim language into plain next steps.	Member Dashboard
Must	Provider Summary Screen	Shows the provider path and turns the solution into a two-sided workflow.	Provider Action Center
Must	Representative Summary Screen	Shows how unavoidable calls become faster and easier to resolve.	Representative View
Should	ROI Status Widget	Shows use of roi_authorizations and prevents dead-end calls.	Member Dashboard / ROI Screen

Priority	Feature	Why it matters	Screen
Should	Care Gap Widget	Shows proactive health support beyond claims.	Member Dashboard / Care Gap Screen
Should	Agent Activity Feed	Makes the multi-agent architecture visible instead of hidden behind one chatbot response.	Side Panel

Copy/Paste Pitch

We proactively identify claim, ROI, and care-gap issues, explain them through an Amazon-style member journey dashboard, generate provider action summaries, and equip representatives with AI-generated context when escalation is required. The top half of the experience prevents unnecessary contacts through visibility and guidance. The bottom half improves the contacts that still happen by giving representatives the root cause, actions taken, and next best step before the call begins.